

Group-A (Answer all questions)

10 × 1 = 10

1. What is the first step in the software development lifecycle? [CO2, Remembering]
☒ a) System Design b) Coding
☐ c) System Testing d) Preliminary Investigation and Analysis
2. What does a directed arc in a DFD signify? [CO6, Understanding]
☒ a) Data Flow b) Data Process
☐ c) Data Stores d) None of the above
3. System Testing performed by a friendly set of customers is called [CO7, Understanding]
☐ a) Alpha Testing ☒ b) Beta Testing
☐ c) Performance Testing d) Usability Testing
4. _____ originally proposed the spiral model. [CO2, Remembering]
☐ a) Pressman ☒ b) Royce
☐ c) IBM d) Boehm
5. Big Bang Integration testing is useful for projects with [CO6, Understanding]
☐ a) smaller number of modules ☒ b) large number of modules
☐ c) average number of modules d) None of these
6. Cyclomatic complexity is used to measure: [CO7, Understanding]
☐ a) The execution speed of a program ☒ b) The number of test cases required for basis path testing
☐ c) The memory usage of an algorithm d) The number of syntax errors in a program
7. To achieve a good design, modules should have [CO6, Analyzing]
☐ a) Weak cohesion Low coupling b) Weak cohesion High coupling
☒ c) Strong cohesion Low coupling d) Strong cohesion High coupling.
8. Equivalence class partitioning is followed in the [CO5, Understanding]
☐ a) white-box testing ☒ b) black-box testing
☐ c) verification d) none of these.
9. Which is NOT a size measure for software? [CO7, Understanding]
☐ a) LOC b) Function count
☒ c) Cyclomatic complexity d) Halstead's program length
10. According to intermediate COCOMO model, number of cost drivers is [CO7, Remembering]
☐ a) 10 ☒ b) 15
☐ c) 20 d) 14

Group B (Answer any five questions)

5X6=30

1. Explain briefly Software development lifecycle. [CO2, Understanding]
2. Differentiate Black Box Testing and White Box Testing. [CO4, Understanding]

3. Draw a labelled DFD for a Library management system. Clearly show the context diagram and its hierarchical decompositions up to level 2. [CO3 Evaluating]
4. Distinguish between software verification and software validation. When, during the software lifecycle are verification and validation performed? [4+2] [CO4, Understanding]
5. Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded in basic COCOMO model. [CO1, Understanding]
6. What are the relative advantages of using LOC or the function point metric to measure the size of a software product? [CO2, Understanding]

Group C (Answer any four questions)

4X10=40

1. a) A software project with the following functional units: User Inputs (60, weight factor=4), user outputs (50, weight factor=5), user enquiries (45, weight factor=4), user files (07, weight factor=10), external interfaces (04, weight factor=7). The Degree of influence of the 14 adjustment factors is average (i.e 3). Compute the FP of the project. [CO3, Evaluating]

b) Explain Delphi Cost Estimation technique. What are its advantages and disadvantages? [5+5]

2. a) What is cohesion and coupling in the context of software design?

b) Enumerate the different types of cohesion that a module might exhibit. [5+5] [CO2 Understanding]

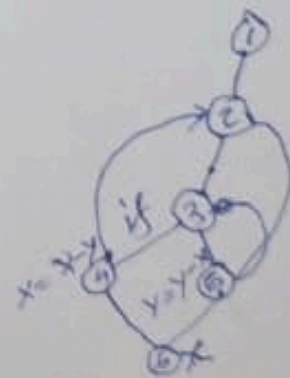
3. a) Explain alpha and beta testing.

b) Differentiate and explain different integration testing approaches.

c) Draw the CFG for the following program piece and calculate cyclomatic complexity using all the methods. List independent paths.

```
int compute_gcd(x,y)
int x,y;
{
while (x!=y)
if (x>y) then
x=x-y;
else
y=y-x;
return x;
}
```

```
while (x!=y) {
if (x>y) {
x=x-y;
}
else {
y=y-x;
}
return x;
}
```



[2+3+5] [CO7, Evaluating]

4. a) What are the properties of a good SRS document? [CO4, Remembering]

b) Explain functional and non-functional requirements. [CO4, Understanding]

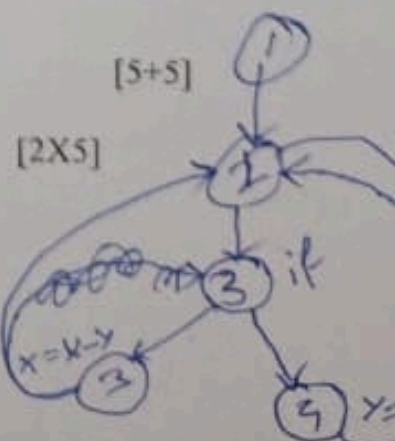
[5+5]

5. Write short notes on: (Any two)

a) Feasibility Study [CO1, Understanding]

b) Spiral Model [CO2, Understanding]

c) Performance Testing [CO7, Understanding]



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